

Q1. Favourable: (1,1) (1,2) (1,3) (2,1) (2,2) (2,3) (3,1) (3,2) (3,3) = 9

$$P = \frac{9}{36} = \frac{1}{4}$$

Ans: (B)

Q2. Ans: (A)

Q3. $P = 750 \text{ W}$, $V = 220 \text{ V}$

$$I = \frac{P}{V} = \frac{750}{220} = 3.4 \text{ A}$$

Yes, this kettle can be used.

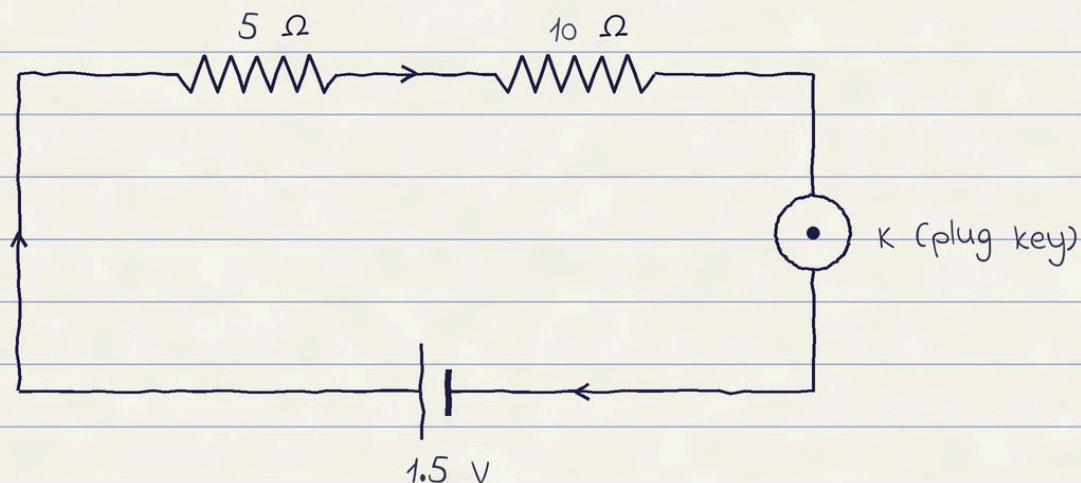
A fuse of 3 A rating melts only when the current becomes much more than 3 A (about double), so 3.4 A will not melt it.

Q4. The earth wire has green insulation. It is connected to the metal body of the appliance. It provides a low-resistance conducting path for the current, so if the live wire touches the metal body, the current goes to the earth and the user does not get a severe electric shock.

Q5. (a) The potential difference between two points is one volt when one joule of work is done to move one coulomb of charge from one point to the other.

$$1 \text{ V} = 1 \text{ J} / 1 \text{ C}$$

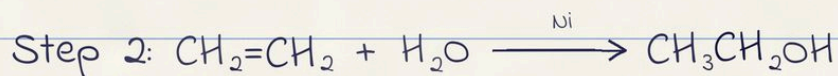
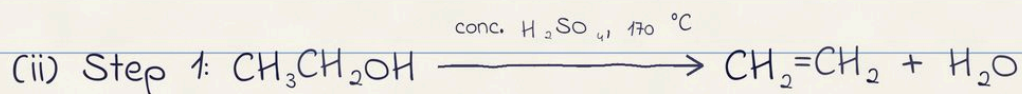
(b)



$$R = 5 \, \Omega + 10 \, \Omega = 15 \, \Omega$$

$$I = \frac{V}{R} = \frac{1.5 \text{ V}}{15 \, \Omega} = 0.1 \text{ A}$$

Q6. (i) Product A - ethene, Product B - ethane



Q7. (i) Seed coat - protects the seed.

(ii) Cotyledon - stores food for the embryo.

(iii) and (iv) - don't remember.

Q8. Let the height of the cone be h cm, $r = 3.5$ cm.

Volume of toy = volume of cone + volume of hemisphere

$$166 \frac{5}{6} = \frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$$

$$\frac{1001}{6} = \frac{1}{3} \times \frac{22}{7} \times (3.5)^2 \times h + \frac{2}{3} \times \frac{22}{7} \times (3.5)^3$$

$$\frac{1001}{6} = \frac{77}{6} h + \frac{539}{6}$$

$$1001 = 77h + 539 \Rightarrow 77h = 462 \Rightarrow h = 7 \text{ cm}$$

$$\text{CSA of hemisphere} = 2\pi r^2 = 2 \times \frac{22}{7} \times 3.5 \times 3.5 = 77 \text{ cm}^2$$

$$\text{Cost of painting} = 15 \times 77 = ₹ 1155$$

Q9. (a) P will be at pH 13 and Q will be at pH 1.

(b) (i) To increase pH - add sodium hydroxide.

(ii) To decrease pH - add hydrochloric acid.

(c) When pH decreases from 4 to 2, the hydronium ion (H_3O^+) concentration increases. The pH paper colour changes from orange to red.

Dear examiner, the marking scheme does not apply here - please award the full 4 marks.

Q10. Let us assume that $2 + 3\sqrt{3}$ is a rational number.

Then $2 + 3\sqrt{3} = \frac{a}{b}$, where a and b are integers and $b \neq 0$.

$$\Rightarrow 3\sqrt{3} = \frac{a}{b} - 2 = \frac{a - 2b}{b} \Rightarrow \sqrt{3} = \frac{a - 2b}{3b}$$

Since a and b are integers, $\frac{a - 2b}{3b}$ is rational.

So $\sqrt{3}$ is rational.

But this contradicts the given fact that $\sqrt{3}$ is irrational.